MOTIONINK

IMU handwriting pen

- Overview
- Live capture
- Transcription
- Event log
- Sessions
- Settings

xG26 / ICM-40627

● Connected over BLE

Local processing

01 / Live capture

Six-axis sensor readings and on-device character detection

● Connected Demo data

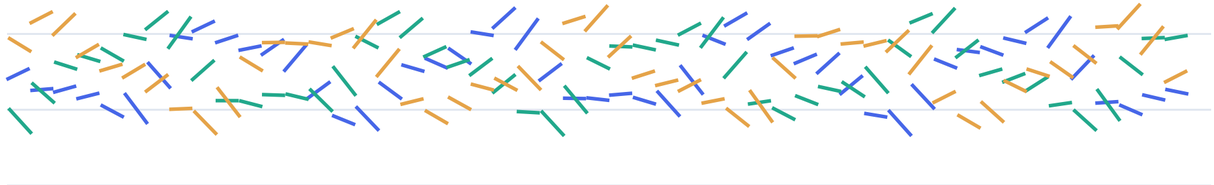
Capture session

Paper · fixed pen mount · separated block characters

● Ready Start capture Stop

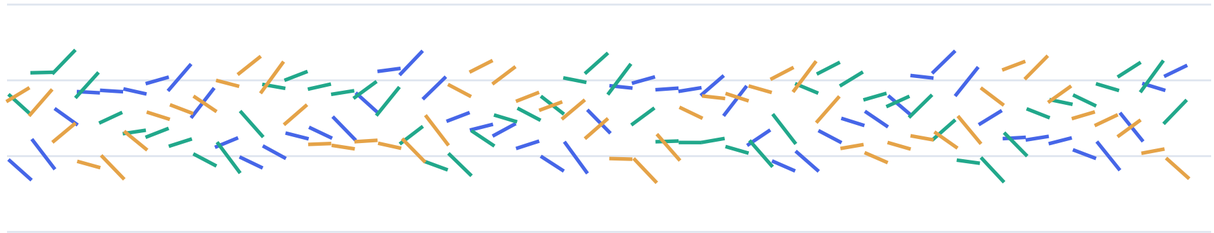
Accelerometer

Acceleration (m/s²) • X, Y, Z X Y Z



Gyroscope

Angular velocity (°/s) • X, Y, Z X Y Z



Latest recognized character

A

Confidence 96.2%

Device diagnostics

Capture state	**Waiting**
Last event	**00042**
Latency	**182 ms**
BLE receipt	**Received**
Delimiter	**Pause or button**

Sensor charts are a debugging view; inference is performed by the embedded TinyML classifier.

02 / Transcription

Keep raw predictions visible while allowing manual corrections

Connected | Demo data

Digital transcription

Session 04

RAW PREDICTIONS • READ ONLY

H E [?] L O

EDITABLE OUTPUT

HELLO

Edit / copy / export

Selected event

Needs review

?

Event #00039

Below confidence threshold

Replace prediction | Reject

Recognition event history

Unmodified classifications retained for evaluation and debugging

EVENT	TIMESTAMP	RAW CLASS	CONFIDENCE	STATUS
00037	14:02:06	H	97.1%	Accepted
00038	14:02:07	E	94.6%	Accepted
00039	14:02:08	?	42.8%	Rejected
00040	14:02:09	L	95.8%	Accepted
00041	14:02:10	O	92.3%	Accepted

03 / Setup & evaluation

Configure the writing protocol and inspect prototype test results

Connected
Demo data

Connect the IMU pen

Silicon Laboratories xG26 Development Kit

EFR32MG26 + ICM-40627 six-axis IMU

Bluetooth Low Energy
Connected

Scan devices
Disconnect

Writing protocol

01 Secure sensor at a fixed pen orientation

02 Choose paper and supported block characters

03 Write one separated character at a time

04 Pause or use the documented boundary control

Evaluation snapshot

Illustrative values shown for layout only • targets from project definition

Accuracy of accepted	Segmentation	95th percentile latency	Rejection rate
94.1%	**91.7%**	**310 ms**	**5.2%**
Target ≥ 90%	Target ≥ 90%	Target ≤ 500 ms	Must be reported

Data collection & export

Keep timestamped IMU samples, event identifiers, original labels, corrections and rejected events locally.

Export IMU CSV
Export event log
Clear session